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Genuine effects of vote advice applications on party choice: Filtering out factors that affect both the advice obtained and the vote

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Abstract
Previous research shows effects of the advice from voting advice applications (VAAs) on party choice. These effects could be spurious because common antecedent factors like prior voting, a voter's prior issue positions and election campaign news may explain both party choice and the opinions someone reports to the VAA, and hence the voting advice obtained from the VAA. Often VAAs will advise users to opt for parties that they were already likely to vote for, based on antecedent factors. Here, three-wave panel surveys and media content data for the Dutch national election campaigns of 2010 and 2012 are employed. In spite of spurious correlations resulting from common antecedent factors, genuine VAA effects show up, especially for doubting voters. Party change based on positive VAA-advice for a party is least likely (a) for voters who already have an abundance of antecedent factors in favour of that party anyway, and (b) for those without a single antecedent factor in favour of that party. Genuine VAA effects imply that VAAs make it less easy for political parties to neglect each other's owned issues, because VAAs weigh issues equally for each party.

Keywords
elections, issue voting, media effects, vote advice applications

An increasing number of voters make their voting decision not until the election campaign has started (Krouwel, 2012; Rose and McAllister, 1986; Weßels et al., 2014). To make an informed choice – based on the relevant issues in the campaign – millions of voters have come to use voting advice applications (VAAs) such as the EUprofiler and EUvox in the 2009 and 2014 European Elections, the Swiss Smartvote, German Wahl-O-Mat and the Dutch Stemwijzer and Kieskompas. VAA websites offer a personalized 'voting advice' based on a match between the issue preferences retrieved from the user and the official positions of parties or candidates on salient political issues. As Garzia, Trechsel, Vassil et.al. (2014) and others argue, VAAs 'take the burden of informing themselves out of the voters' hands and deliver concise, easy to interpret and readily digestible information'.

Scientific interest into the usage and effects of VAAs can be seen from a growing number of research articles and overviews thereof, such as Garzia and Marschall (2012: 368), Garzia et al. (2014) and Rosema et al. (2014). Starting from different election campaigns, different methods and different samples, most studies found that VAAs enhance interest in the political campaign and increase political

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knowledge (Kamoen et al., 2015) as well as turnout (Dinas et al., 2014; Ladner and Pianzola, 2010; Marschall and Schultze, 2012; Pianzola, 2014b; Ruusuvirta and Rosema, 2009) with some exceptions (Enyedi, 2015). The obtained advice to vote for a specific party appears to affect the vote. The number of studies that found a substantial VAA influence on the likelihood of voting for the advised party (Kleinnijenhuis et al., 2007a; Ladner et al., 2012; Marschall and Schultze, 2012; Pianzola, 2014a, 2014b; Ruusuvirta and Rosema, 2009; Wall et al., 2014) exceeds the number of studies that did not (Enyedi, 2015; Kamoen et al., 2015; Walgrave et al., 2008).

The vast majority of studies that showed that issue-based advice from a VAA affected the vote, however, did neglect the funnel of causality. The authors of *The American Voter* (Campbell et al., 1960) showed that issue concerns immediately affect vote choice for some voters, but that these opinions on issues are partly shaped by antecedent factors culminating in a long-standing party attachment, and by newsworthy events in the unfolding campaign. The studies that showed an effect of issue-based advice from a VAA on the vote did, however, not take into account how party attachment could affect the opinions voters have on issues covered by VAAs, which subsequently affect the VAA advice. Additionally, these previous studies also neglected effects of information and communication, like media influence.

In this study, we seek to isolate the effect of advice obtained from a VAA on vote choice from the effects of antecedent factors that produce a spurious correlation between advice from a VAA and the vote. A spurious correlation between VAA advice and the vote arises when voters 'self-select' the advice they obtain by feeding the VAA with opinions that would have driven them anyhow to the party advised by the VAA (Levendusky, 2011; Pianzola, 2014a, 2014b). As Andreadis and Wall (2014) note, we need to study 'the factors that exacerbate or minimise' the impact of VAAs. Our *research question* reads: *Does advice from a VAA exert an influence on vote choice in addition to other, antecedent factors, when the dependency of VAA advice on these antecedent factors is taken into account*?

Why advice from a VAA may affect the vote

As Garzia (2010) and others argue, autonomous VAA effects on the vote may appear because VAAs offer voters a convenient shortcut to find out about all party positions on pressing issues. As compared to voters' spontaneous agreement on the issues with parties, VAAs include more and different issues than voters originally had in mind, including other issues than the issues on which specific parties focused in their campaign. The issues that are included in a VAA bear the same weight for each party, although voters are inclined to judge each party on salient party-specific issues in short on 'owned' issues (Budge and Farlie, 1983). Only a minority of voters instruct a VAA to not consider issues that they would not have considered spontaneously. The new information that users of VAAs receive may also draw attention to previously unknown parties (Garzia et al., 2012). Other reasons why VAAs may have an additional effect on the vote is that the VAA result is not only dependent on users' opinions but also on design aspects such as the selection of issue statements (Lefevere and Walgrave, 2014; Walgrave et al., 2009), specific statement wordings (Holleman et al., 2016), and on a specific choice from a variety of matching algorithms (Louwerse and Rosema, 2014; Otjes and Louwerse, 2014).

The core expectation on the basis of this literature is that the obtained advice from a VAA to vote for a party adds to the explanation of vote choice (hypothesis 1a), whereas VAA use resulting in a 'negative' advice, that is, to vote for another party, decreases the likelihood to vote for this party (hypothesis 1b). In many elections, voters may consult more than one VAA because more VAAs are available. Inconsistent voting advice from different VAAs may reduce the likelihood that the advice will be followed (Enyedi, 2015) because their inconsistency may prompt users to resort to other information (Israel et al., 2016). These expectations are summarized in hypothesis 1.

> **H1:** VAA advice to vote for a party increases the likelihood to vote for this party at the upcoming elections (hypothesis 1a); negative advice (hypothesis 1b) and inconsistent advice (hypothesis 1c) decrease this likelihood.

Antecedents that affect both VAA advice and the vote

The question how to isolate the genuine effect of advice from VAAs on vote choice starts from theories about antecedent factors that may affect both the VAA advice obtained and the ultimate vote choice. Starting from *The American Voter* (1960), many studies showed that the vote is embedded in a historical record of party identification and party attachment, which in turn may be based on long-standing value orientations, political socialization and dominant social divisions. Since party identification/attachment is not a very distinctive concept in a multiparty systems (Berglund et al., 2005; Thomassen, 1976), we will consider previous party choice instead. Our twofold first hypothesis therefore reads:

> **H2**: Having voted for a party at the preceding elections increases the likelihood to vote for this party at the upcoming elections (hypothesis 2a) and the likelihood to obtain VAA advice to vote for this party (hypothesis 2b).

Many voters change party preferences already before the next election campaign (Rattinger and Wiegand, 2014), which is also obvious from the electoral outcomes of many midterm elections. Since one's party preference at the start of the new election campaign is a more proximate cause in the funnel of causality than prior party choice, it is to be expected that party preference at the start of the election campaign also affects VAA advice and party choice.

H3: The intention to vote for a party at the start of an election campaign increases the likelihood to vote for this party at the upcoming elections (hypothesis 3a) and the likelihood to obtain VAA advice to vote for this party (hypothesis 3b).

In the final months leading up to the election, voters respond strongly to the issues and the candidates put forward by the parties. Even voters who shift to another party in a multiparty system usually shift to a party with similar issue positions (Dassonneville and Dejaeghere, 2014; Van der Meer et al., 2015). We expect therefore that issue preferences at the start of the campaign affect the personal opinions fed into a VAA, and consequently the VAA advice obtained, and ultimately one's final party choice. Spontaneous issue agreement is conceptualized here as the extent to which a voter agrees with a party on salient issues they associate with that party. According to the theory of issue ownership (Budge and Farlie, 1983; Petrocik, 1996; Walgrave et al., 2012), historical social cleavages (e.g. rural vs. urban, poor vs. rich) gave rise to the birth of parties that thereafter consistently promoted specific interests with which they are associated by voters.

H4: Agreement on the issues with a party at the start of the election campaign increases the likelihood to vote for this party at the upcoming elections (hypothesis 4a) and to obtain VAA advice to vote for this party (hypothesis 4b).

Voters react to context variation. Which party they vote for, whether VAAs are available, whether voters make use of VAAs, and with which preferences they feed VAAs, depends on the prevailing political system, the party landscape, the candidates and on which party gains momentum during the campaign. This study is based on a most similar design with two subsequent parliamentary elections in 2010 and 2012 in the Netherlands. In 2010 and 2012, the same three parties competed: the liberal party VVD, lead by Mark Rutte, the most important challenger, the PVV, by Geert Wilders, and the second largest party, the Labour Party (PvdA). To capture the remaining contextual variation between elections and between parties, we will focus on 'horse race news' (Banducci and Hanretty, 2014; Mutz, 1997; Patterson, 1993).

Horse race news in the media reflects impressions of momentum in the polls, debates and public appearances. It signals which candidate is considered most competent to win the elections and to rule the nation. Previous research shows that news about successes and failures of parties exerts a stronger effect on party choice than news about the issue positions of parties (which fosters prospective voting), about real-world cues and issue developments (which fosters retrospective voting) and attack news (which fosters voting based on social identification and cognitive balance) (Kleinnijenhuis et al., 2007b). Horse race news attracts news consumers (Iyengar et al., 2003) and directs their attention also to other election messages, including issue information (Bleske and Zhao, 1998). In line with Walgrave et al. (2008), we expect that the effect of a VAA is strong when it confirms general trends, when it reinforces the winner and causes a bandwagon effect. This leads to our fifth hypothesis:

H5: Exposure to horse race news (news about success and failure) during the election campaign in favour of a party increases the likelihood to vote for this party at the upcoming elections (hypothesis 5a) and to obtain VAA advice to vote for this party (hypothesis 5b).

The user perspective

The impact of VAA advice on the actual party choice will probably not be the same for each voter. A study of use patterns showed that VAAs are used by different types of users for different reasons (van de Pol et al., 2014). The largest group of users can be characterized as politically sophisticated voters who are relatively certain about their vote choice. VAAs are mainly used by them to affirm that their views correspond with those of their favourite party. Van de Pol et al. (2014) labelled this type of users 'checkers'. Since checkers very often already have decided which party to vote for independent of VAAs, we expect that checkers change their vote choice least often in accordance with VAA advice. VAAs were also found to be consulted by a, somewhat smaller, group of users who could be described as possessing less political sophistication and interest, and as being very uncertain on which party to vote for. This group could further be subdivided into 'seekers', a rather efficacious but uncertain group (they put a lot of importance on the voting advice they receive) and 'doubters', who are much less efficacious and more cynical about their possibilities of affecting political decision-making. We expect the seekers to be most affected, since they are still undecided but motivated to make the 'right' decision.

H6: As compared to checkers (the baseline), seekers (hypothesis 6a) most often bring their vote choice in line with VAA advice, followed by doubters (hypothesis 6b).

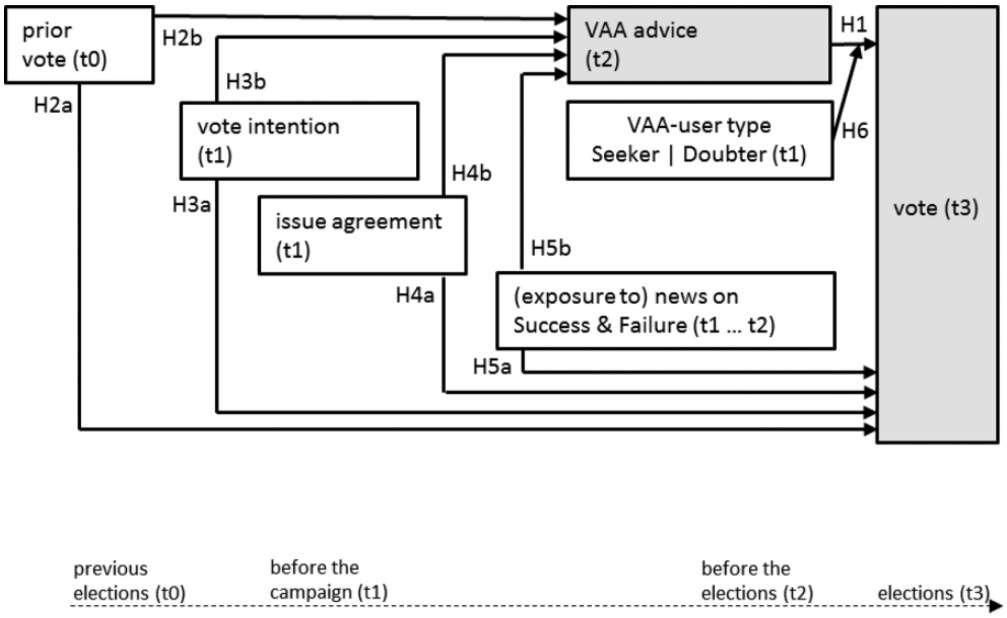


Figure 1. Conceptual model.

Towards a conceptual model

Figure 1 combines the hypotheses in a conceptual model and shows in addition in which of the three waves the variables in the model are measured.

Figure 1 shows how the research model predicts the effect of VAA advice on the vote (hypothesis 1) on top of the vote at the previous election (hypothesis 2), prior vote intentions (hypothesis 3), prior issue agreement (hypothesis 4) and news about successes and failures of parties (hypothesis 5) and how these independent variables (considerations) are placed in time. The effect of VAAs is moderated by the nature of VAA users (hypothesis 6): seekers and doubters, with checkers as the reference category.

Method

Data

The aim of this study is to isolate genuine VAA effects on vote choice when both the obtained VAA advice and vote choice depend also on common antecedent factors. A three-wave panel survey study with a wave before the campaign, a wave before the elections and a wave just after the elections (Huckfeldt and Sprague, 1995) allows for the detection of self-selection that occurs when a voter with specific predispositions at time 1 reverts to these predispositions when answering VAA questions at time 2, which results in a vote choice at time 3 that is both consistent with predispositions and the obtained VAA advice. We label these three waves as the pre-campaign wave, the pre-election wave and the post-election wave, or alternatively, as wave 1, wave 2 and wave 3.

Three-wave panel survey data also allow for the distinction between *potential effects* of VAA-advice, like in experimental research, and *genuine effects* that take into account the actual use of VAAs and the actual correspondence of VAA advice with a voter's prior considerations. Panel data are worthwhile furthermore to re-check to what extent effects of self-reported VAA-advice persist until after the elections (Hanel and Schultze, 2014). Recording the self-reports about VAA use and the VAA advice at time 2, just before the elections, makes it unlikely that VAA advice is forgotten already and also unlikely that these self-reports are fabricated to justify one's vote choice, since the latter is recorded at time 3, after the elections.

The polling company GfK conducted such panel surveys both for the 2010 and 2012 elections in the Netherlands in line with ESOMAR and ISO 26362 guidelines for representative computer-assisted web interviewing panels. The respondents in the first wave of these panel survey studies represent a stratified sample of a large multi-access panel (MAP) of respondents (*N* about 100,000). The MAP rests on multiple recruitment strategies to enable large-scale samples that match population characteristics of the Dutch population (95% Internet penetration) including a not-so-heavy internet use on the average. About 31% of MAP respondents were recruited by a targeted search in various difficult to reach subgroups of the population. Stratification for the first wave of the 2010 and 2012 three-wave panel survey studies was based on turnout and voting choice at the previous election, age, education, gender and Nielsen regions. The stratification guaranteed that difficult to reach groups such as non-voters, new voters and voters for parties

with many less educated voters were proportionally represented in the 2010 and 2012 election campaign panel survey studies. The number of respondents in the first waves of the two panel studies amounted to $N = 1801$ and $N = 1805$, respectively.

The first wave was administered in a weekend at the start of the election campaigns leading up to the national parliamentary elections of 2010 and 2012 (first weekend of April 2010/of July 2012) and contained questions on media exposure, political knowledge, a retrospective question on the vote cast at the previous elections and the party preference at this stage. The second wave, carried out in the 4 days immediately before the elections (second weekend of June 2010/of September 2012), captures whether or not respondents used one or multiple VAAs and, if they did, what was the voting advice provided by the VAAs. The third wave, administered after Election Day (9 June 2010/ 12 September 2012), assessed whether the respondent voted, and if so, for which party. Reminder emails were sent to increase the response rate.

The analyses are based on the subset of respondents who participated not only in the pre-campaign wave but also in the pre-election wave and the post-election wave of the panel survey studies. The final numbers of respondents in the analyses amount to $N = 1159$ in 2010 and $N = 1243$ in 2012. Since panel attrition was only weakly correlated with party choice (measured in the post-election wave)[1], respondents who participated in one or two waves only could be excluded from the study without having to resort to data imputation methods or to analyses based on varying numbers of respondents.

The analyses are based on the parties that managed to gain seats in the Dutch parliament with its electoral threshold of 0.67% only: GroenLinks (ecologists), Partij voor de Dieren (animal rights), SP (socialists), PvdA (Social-Democrats), D66 (liberal democrats), CDA (Christian Democrats), CU (orthodox Christians), SGP (orthodox Christians), VVD (right-wing liberals) and PVV (anti-immigration) both in 2010 and 2012 50+ (elderly) was added in 2012. A new party, 50+ (elderly), was added in 2012.

Measures

Vote at previous elections. In the pre-election wave, respondents were asked which party they had voted for at the previous election. For respondents who had already participated in GfK research, the vote at the previous elections was based on the earliest measurement.

Vote intention at the start of the campaign. In the pre-campaign wave, voters were asked whether they intended to cast their vote, and if so, they were asked: 'For which party would you vote if elections were held today?' with the parties as answer options.

Vote. In the post-election wave, respondents were asked whether they had voted at the last election. If they did, the follow-up question asked which party they had voted for.

Voting advice applications. In the second wave, voters was asked whether they had used the most popular VAA in the Netherlands, that is, whether they had used Stemwijzer and whether they had used Kieskompas. Voters who had used one or both applications also answered the question: 'What was the advice you obtained from [VAA]?' with the parties as answer options. For each combination of voters and parties, three dichotomous variables were construed to assess VAA-effects. The first indicates whether a party was recommended by any VAA: either two VAAs were used which all recommended voting for this party or only one VAA was used that recommended this party. The second variable indicates whether all VAAs that were used advised another party than the party at hand. The third variable indicates whether voters received inconsistent advice: one VAA recommended the party and the other VAA did not.

Issue agreement. In the pre-election wave, respondents were asked which issues came to mind when considering each of the parties. Issues could be listed freely, but 97% of the respondents chose issues from a list of suggestions. For each party, two different issues could be selected. Next, voters were asked whether they disagreed completely (−1), disagreed (−0,5), nor agreed nor disagreed (0), agreed (0,5) or agreed completely (1) with that party with respect to these issues (Kleinnijenhuis and Pennings, 2001). Issue agreement was computed as summated issue agreement per issue.

Tone of success and failure news. A content analysis of all political news items from five national newspapers, two free dailies and two television news programmes (one from a public and one from a commercial broadcaster) was conducted to measure the tone of the news. The tone of this news was indicated for each specific medium by the number of statements about successes minus the number of statements about failures during the entire election campaign. This type of news deals with performance attributions, for example on the basis of policy failures and successes, or of performance in debates, in the polls, or in television programs. In the relatively long 2010 campaign, $n = 2732$ statements in the investigated media dealt with party successes or failures, as compared to $n = 1924$ in the shorter post-summer 2012 campaign. Experienced student coders coded the weekly news during each week of the election campaigns. This resulted in sufficiently reliable coding results as measured by Krippendorff's α ($\alpha = 0.79$ for 2010 and $\alpha = 0.92$ for 2012). The tone of the success or failure news in a medium was assigned only to its readers or watchers (e.g. Boomgaarden et al., 2011; Kleinnijenhuis et al., 2007a). Therefore, respondents were asked which

specific newspaper and television news programmes they used during the last week, and how often. About 90% of the respondents stated to have read or watched one of the investigated media during the last week at least once. Success and failure for a specific party in the news was measured by the number of statements about successes minus the number of statements about failures during the entire election campaign in the media that were actually used by a respondent. Voters who followed none of the investigated media were assigned a neutral score, which reflects that they did neither obtain news about successes nor about failures of parties. To obtain an independent variable for which logistic regression coefficients could be interpreted, the resulting variable was standardized by taking *z*-values. Social media are not included, because we do not dispose of measures of the political Facebook or Twitter content followed by individual users. Effects of social media on party choice should not be exaggerated since even in 2017 less than 20%, even of the youngest category of voters, obtained political information through social media and less than 5% exclusively through social media.

Type of VAA user. The categorization of voters into checkers, seekers and doubters is based on data from the first wave. *Checkers* were defined as the reference category of respondents who intended to vote for the same party they had voted for at the previous national elections and did not also consider voting for another party. The latter is determined by comparing the respondents' propensity to vote for this party with that of all other parties. *Seekers* and *doubters* were defined as non-checkers; seekers were further distinguished from doubters by relatively high scores on the weighted average of news exposure (weight = 1) and political interest (weight = 1) and a relatively low score on cynicism (weight = 2). Cynicism was weighted double because it acts as a single counter weight against political interest and news exposure that tap more or less the same concept. The cut-off point on the combined political interest–news exposure–cynicism scale to distinguish seekers from doubters was brought in line with the 3:1 ratio of seekers versus doubters in previous research (van de Pol et al., 2014). The underlying variable *news exposure* was measured using the question how often the respondent had followed the news during the previous week. *Political interest* was measured using three items in the pre-campaign wave. Cronbach's α for 2010 amounts to 0.78 and for 2012 to 0.77.[2] *Political cynicism* was measured using five items in the pre-campaign wave.[3] Cronbach's α for 2010 amounts to 0.78 and for 2012 to 0.80.

Data analysis

A multilevel random intercepts logistic regression model was estimated in order to analyse the effect of one's previous vote, one's vote intention at the start of the campaign, the tone of success and failure news during the campaign, one's issue agreement with parties and the voting advice provided by VAAs as measured shortly before elections, on the final vote choice. In this model, parties are nested within individuals. The variables in the two multilevel regression equations vary both between respondents and between parties, except for the seeker and doubter variables, which vary between respondents only. The intercept in the multilevel regression models is allowed to vary between parties.[4] Since the estimated model is basically an autoregression equation in which the vote depends on the previous vote, the estimated effects of the other variables can be interpreted as dynamic effects on party *change* rather than as static effects on party choice.

Multilevel logistic regression coefficients do however provide limited information about actual effect sizes (Mughan, 2015). The regression estimates give at best an indication of *potential* VAA effects. Therefore, we include an additional analysis to assess *genuine* VAA effects. Genuine effects differ from potential effects because they do not assume that a stimulus, like an advice to vote for a specific party, is administered to 100% of the respondents in a given experimental condition. Rather they take into account the probability that the stimulus was actually administered. A VAA is typically not used by all voters and typically does not recommend a specific party to all voters. The data and the syntax files used are permanently available from https://dataverse.nl/dataset.xhtml?persistentId=hdl:10411/77APHO.

Results

VAA users and VAA impact

Table 1 presents elementary data on the overall usage of VAAs and the aggregate relation between VAA advice and the final vote by focusing on the differences between the three types of VAA users: checkers, seekers and doubters.

The table shows that Stemwijzer is more often used than Kieskompas. Seekers use Stemwijzer and Kieskompas more often than checkers, who in turn use them more often than doubters.

The last two columns of Table 1 refer to those who voted in line with the VAA advice in the final count as a percentage of all VAA users who voted against their preference at the start of the campaign. The percentages show that doubters, who were least inclined to use a VAA, seem to be most susceptible to their advice, once they decide to use a VAA. Seekers, who were most inclined to use a VAA, were less susceptible to VAA advice than doubters, but still more susceptible than checkers. These differences between seekers and doubters show that very different motives play a role in using a VAA and following the advice from a VAA.

Table 1. VAA use and VAA impact for checkers, seekers and doubters.*

		VAA user type in year	VAA users as a % of all respondents		VAA users who vote in line with the VAA advice obtained as a % of VAA users who received an advice that was different from their preference at the start of the campaign[1]	
VAA user type	Year	%	Stemwijzer	Kieskompas	Stemwijzer	Kieskompas
Checkers	2010	33	32	12	13	9
	2012	23	35	13	11	12
Seekers	2010	47	45	18	27	20
	2012	42	46	16	28	17
Doubters	2010	20	30	8	34	20
	2012	35	32	9	34	14

Note: VAA: voting advice applications.
*Number of respondents: 2010 $n = 1159$; 2012 $n = 1243$. Stemwijzer users 2010: $n = 313$; 2012 $n = 344$. Kieskompas users 2010: $n = 113$; 2012: $n = 119$. *F*-tests (not reported here) show that differences due to VAA user type are significant.

Table 2. Random intercept models to explain the vote.

		Obtained VAA advice				Vote choice					
		empty model				empty model					
Dependent variable		(1a)		(1b)		(2a)		(2b)		(2c)	
H2	Prior vote			0.98	***			1.98	***	1.90	***
H3	Pre-campaign vote intention			0.93	***			2.93	***	2.93	***
H4	Prior issue agreement			0.55	***			0.59	***	0.57	***
H5	News success/failure			0.03	*ns*			0.10	**	0.10	**
H1	VAA advice										
	H1a Advised by VAA									1.09	***
	H1b Not advised by any VAA									−0.25	***
	H1c Inconstency VAAs									−0.59	*
	H6a Advised by VAA to a seeker									0.74	**
	H6b Advised by VAA to a doubter									1.33	***
Intercept		−2.58	***	−2.95	***	−3.04	***	−4.07	***	−4.08	***
Variance of intercept across parties ($n = 11$)		0.93		0.67		2.05		0.96		0.84	
n cross-nested											
Year-party-respondent observations		10,092		10,092		25,263		25,263		25,263	
Goodness-of-fit											
	AIC	6020		5298		13,158		7612		7348	
	DIC	5907		5187		13,018		7491		7220	

Note: VAA: voting advice applications; AIC: Akaike information criterion; DIC: deviance information criterion.
*$p < 0.05$; **$p < 0.01$; ***$p < 0.001$.

Model test

Table 2 shows estimates for regression models to explain VAA advice for a party (models 1a, 1b) along with estimates for models to explain the vote for a party (models 2a, 2b, 2c). The number of observations for models 1a and 1b ($n = 10{,}092$) is lower than for models 2a, 2b and 2c ($n = 25{,}263$) because only respondents who used a VAA are included in models 1a and 1b.

Model 1b from Table 2 explains the obtained VAA advice to vote for a specific party from structural characteristics, notably from one's prior vote (hypothesis 2b), pre-campaign vote intention (hypothesis 3b), prior issue agreement (hypothesis 4b) and news about party successes in one's media (hypothesis 5b). The regression estimates show that the obtained VAA advice follows from one's prior vote, one's pre-campaign vote intention and prior issue agreement with a party, but VAA advice is not affected by news about successes and failures. The good news here is that one's political predispositions are at the heart of the obtained advice from a VAA, but that this advice is independent of the attributed momentum to political parties in the media.

Model 2b shows that the vote for a specific party is influenced by the same three structural characteristics as

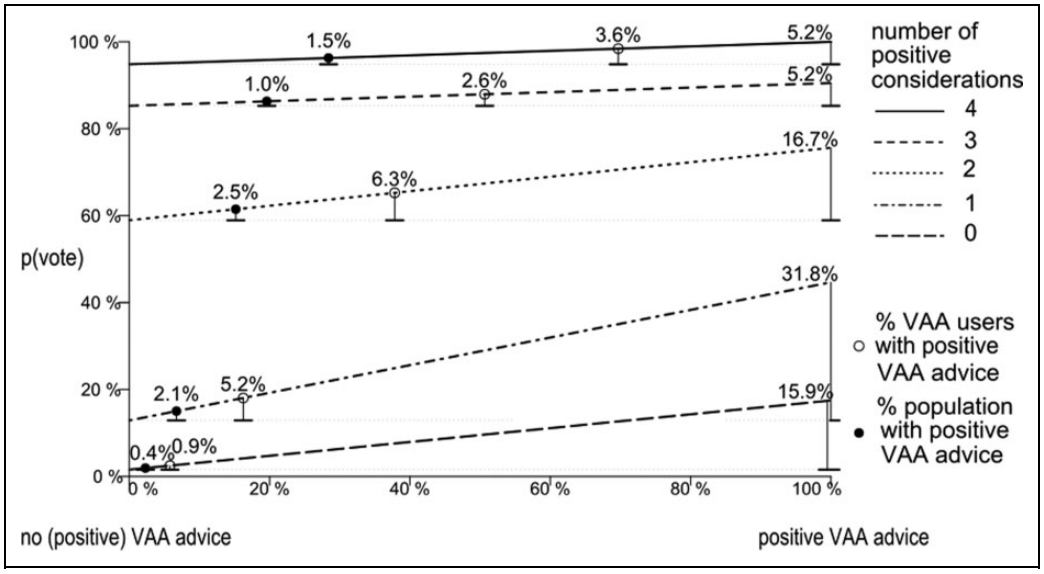


Figure 2. Effect of party considerations on VAA advice and effects of VAA advice. The slopes of the five lines show how the effect of VAA advice to choose a party shortly before the elections (*x*-axis) on the probability to vote for that party at the elections (*y*-axis) is moderated by the number of prior positive considerations to vote for that party. The potential effect of obtaining VAA advice to vote for a specific party for a voter with only one positive considerations to vote for that party amounts to 31.8%. The genuine effect is limited to 5.2% (blank dot) if the fairly low probability is taken into account that a voter with only one positive considerations regarding a specific party would receive such VAA advice and to 2.1% (black dot) only if it is taken into account moreover that the majority of voters do not use a VAA. VAA: voting advice applications.

VAA advice, but unlike the obtained VAA advice party choice depends also in part on news about party successes and failures according to the news. Although the same structural determinants affect both VAA advice (model 1b) and party choice (model 2b), model 2c shows that the obtained VAA advice adds significantly to the explanation of party choice. Voting for a party becomes more likely if the party was advised by any VAA (hypothesis 1a), but decreases in case the party was not advised by any VAA (hypothesis 1b) and in case of inconsistent advice from different VAAs (hypothesis 1c). The negative coefficient for inconsistency (hypothesis 1c) is less strong than the positive coefficient of VAA advice to vote for a party (hypothesis 1a), which implies that a party benefits from recommendations from a VAA, even if another VAA does not recommend the party. As compared to the effect of VAA advice on party choice for the baseline category of checkers, the effect of VAA advice is stronger for the category of seekers (hypothesis 6a) and especially for the category of doubters (hypothesis 6b).

In short, VAA advice matters for party choice, even when controlled for structural determinants (hypotheses hypothesis 2 to hypothesis 5) that affect both VAA advice and party choice. This conclusion is warranted also by other indicators of model fit. Since lower values of Akaike information criterion (AIC) and deviance information criterion (DIC) indicate a better fit, the final models 1b and 2c are indeed superior to models 1a, respectively, 2a and 2b. The final models explain moreover differences between parties, since extending the empty model with additional factors decreases the variance of the intercept across parties.

Potential effect size and genuine effect size

Regression coefficients from a multilevel logistic regression model are hard to interpret, among others because of their non-linearity. Moreover, they represent *potential effects* that indicate the difference between those who did obtain a specific advice from a VAA and those who did not, which easily leads to an overestimation of effect sizes as compared to the common sense notion of effect sizes. Therefore, Figure 2 presents an analysis of *genuine effects* in addition to potential effects. The genuine effect of a VAA on the probability to switch parties is more modest than its potential effect for two reasons. First, the majority of voters still does not use a VAA. Next, the probability to obtain advice from a VAA to vote for a specific party is lowered by a priori considerations in favour of other parties.

Figure 2 resembles the figures that are common in experimental research to show the *potential* effects of a stimulus for various groups, with on the *x*-axis the distinction between voters who did not and voters who did receive VAA advice to vote for a party. Figure 2 is based on average scores across parties and across elections. The values at 0% represent the potential chance to vote for a party, in the case that none of the voters obtained a VAA advice.

Likewise, the values at 100% represent the potential chance to vote for a party in the case all voters obtained a VAA advice. The five lines in Figure 2 represent the number of positive considerations to vote for a party, apart from VAA-advice. To enable a count of whether voters have 0, 1, 2, 3 or 4 positive considerations to vote for a party, the four considerations according to the multilevel regression results presented in Table 2 are weighted equally heavy in Figure 2: party choice at the previous elections (hypothesis 2), the intention to vote for this party at the start of the campaign (hypothesis 3), agreement with that party on salient issues (hypothesis 4), and news about successes rather than failures of this party in one's media (hypothesis 5). Figure 2 is therefore based on a dichotomization of the continuum of values for issue agreement and news about successes and failure. Figure 2 does not directly account for the difference between checkers, seekers and doubters nor for additional effects of using more than one VAA.

The potential effect of VAA advice is simply the difference between the predicted values at 0% (voter did not use a VAA or the party was not advised by the VAA) and 100% (party advised by VAA).[5] The slopes of the five lines tell already that the potential effect of VAA advice is strongest, for voters with *one* positive consideration to vote for a party: due to positive VAA advice, their likelihood to vote for the party increases with 31.8%, as compared to 15.9%, 16.7%, 5.2% and 5.2% for voters with zero, two, three or four positive considerations.

Let us now turn to the genuine effects that are represented by the percentages associated with the black dots in the figure. Whether potential effects result in *genuine* effects is determined by the actual use of a VAA (40% of the voters) in combination with the likelihood that the VAA gives a positive advice. Only 2.5% of voters without a single positive consideration to vote for a party use a VAA and receive moreover advice to vote for that party, as can be seen from the horizontal position of the black dot on the zero positive considerations line. The corresponding genuine effect of a VAA amounts to 0.4% only, as compared to the potential effect of 15.9%. The percentages at the blank dots show VAA effects that would have occurred if instead of 40% of the voters all voters would have used a VAA, with an unaltered distribution of VAA advice.

Figure 2 shows that genuine effects of VAAs are dampened for two different reasons: without any positive considerations to vote for a party, the likelihood to vote for a party increases only with 0.4% because the likelihood to obtain VAA advice in favour of that party is low. But with more than two positive considerations to vote for a party, a VAA advice in favour of a party adds only 1.0%, respectively, 1.5% to the final vote choice because the majority of these voters will vote for that party even in the absence of VAA advice. Voters with many positive considerations already were defined as the reference category of 'checkers' in the multilevel model in Table 2, for whom VAA effects were smaller than for seekers and doubters. Genuine VAA effects are strongest if a voter holds no less and no more than one or two positive considerations towards a party – for example, prior vote choice, but undecided for which party to vote at the start of the campaign. For these voters, the genuine effect of VAA advice amounts to 2.1%, respectively, 2.5% – or three or four seats in the 150 seats Parliament. Such large effects occur especially for large parties, since having voted for a party at the previous elections is already one positive consideration to vote once more for the same party.

Discussion

This article addressed the potential and genuine effect of an advice from voting advice applications (VAAs) on party choice, while taking the 'self-selection' of advice from a VAA into account. Self-selection of an obtained voting advice from a VAA means that voters with many prior positive considerations to vote for a party are also likely to obtain advice from a VAA to vote for that party. To tackle self-selection effects, the current study employs two representative three-wave panel survey studies in the Netherlands in the context of the 2010 and 2012 parliamentary elections with a pre-campaign wave to tap antecedent variables that affect both the VAA advice obtained and the final party choice, a wave immediately before the elections to tap VAA advice, and a post-election wave to tap party choice.

Multilevel regression analysis shows that both party choice and the advice obtained from a VAA to vote for a specific party depend in part on the same positive antecedent considerations: the party voted for at the previous elections, the party preferred at the start of the election campaign, the party one agreed most with on salient issues at the start of the campaign and the party to which successes were attributed in self-selected media during the campaign – the effect of the latter on the obtained VAA advice was however insignificant. In spite of their common origin in positive antecedent considerations, VAA advice adds nevertheless significantly to an explanation of the final party choice.

Starting from a threefold typology based on previous research (van de Pol et al., 2014), effects were especially strong for 'seekers' and 'doubters' as compared to checkers. Politically sophisticated seekers most often use VAAs, but less sophisticated doubters are influenced most heavily by the VAA advice, whereas checkers tend to vote for their favourite party even in the case of VAA advice in favour of another party. The study shows moreover that advice from a VAA to vote for another party and inconsistent advice from multiple VAAs diminish the probability to vote for a party.

We delved into the question whether potential effects of obtaining the advice to vote for a specific party materialized as genuine effects, that is, whether voters actually used a

VAA and whether they obtained such advice. Genuine effects were especially strong for voters who had enough positive considerations in advance to obtain VAA advice to vote for that party, but not enough positive considerations to vote for that party without additional VAA advice. Similar curvilinear effects have been demonstrated for effects of political news on political behaviour (Zaller, 1992).

The current individual level study on the impact of VAAs on the vote given antecedent factors that affect both the VAA advice obtained and electoral choice did not include factors at the party level and the level of VAA design. This study did not look, for example, at the issues that were addressed by VAAs, although the choice of issues is extremely important in the light of agenda setting and issue ownership theories (Walgrave et al., 2009). The study did not look at the question for which party a VAA was detrimental or beneficial (Alvarez et al., 2014).

The core message of this study is that VAAs exert a small effect both on voters with predispositions that turn them either into strong proponents or into strong opponents of a party, but a strong effect on voters with no more and no less than one or two positive considerations to vote for a party. This result suggests a reason why VAAs may be used more often and may have stronger effects in multiparty systems than in majoritarian two-party systems in which voters trust that they do dispose already over many considerations to vote for one of the two parties VAAs lead to party *switching* especially if their advice is confirmed by other recent cues in the political campaign, for example, by trends in horse race news (Walgrave et al., 2008) as was shown in this study, or – although this was not investigated in the current study – by retrospective news about the performance of governing parties with respect to the issues of the campaign (Fiorina, 1981; Sanders and Gavin, 2004). VAAs make issues important for each of the campaigning parties, beyond owned issues on which they traditionally tend to focus (Budge and Farlie, 1983). VAAs may broaden the political debate by fostering parties to take issue positions on issues on which they would not focus otherwise.

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Notes

1. Panel attrition was 35.6% and 31.1%, respectively. The insignificant correlation coefficients between attrition after the first or second wave with the intention to vote for a specific party as measured in the first wave of the panel survey study shows that panel attrition did not affect the dependent variable in the research model.
2. The three items were: When newspapers report about national news, for example, news about government issues, how often do you read that? If problems in our country are discussed in your presence, do you usually participate in the conversation, do you listen with interest, don't you listen or are you not interested? Are you very interested in political issues, fairly interested, or not interested?
3. The five items were: Against all knowledge politicians promise more than they can deliver.
   Ministers and State Secretaries pursue their own interests above all.
   One becomes an MP through your political friends rather than your abilities.
   Most politicians are competent people who know what they are doing.
   Politicians are able to resolve the most important problems.
4. This analysis is performed on a stacked 'long' data set with a row for each possible combination of respondents, parties and election years, and as dependent variable a binary yes/no-vote variable. An alternative approach would have been to analyse a 'wide' dataset with one row per respondent and party choice as a nominal dependent variable in a multinomial logistic regression. However, a multinomial model also estimates theoretically implausible cross-party influences, such as the extent to which the probability of voting for Labour is influenced by news about successes for the Christian Democrats and a VAA advice to vote for the extreme right. We also tested random slope models with party-dependent slopes for VAA advice effects, but these models did not add significantly to the explanatory power of a random intercept model (as measured by the decrease of AIC and DIC).
5. As an alternative for multivariate regression models to estimate potential effects, balanced entropy weighting has been proposed as a way to simulate an experiment. The idea of balanced entropy weighting is to simulate random assignment to experimental stimuli by reweighting the many respondents in a (panel) survey (Hainmueller, 2012). Applied to this article respondents should be reweighted in such a way that VAA users who obtained VAA advice to vote for a specific party are statistically indiscernible from other voters with regard to the mean values of other independent variables (cf. hypotheses 2 to hypothesis 5) and possibly also with regard to their variance and their mutual correlations. This reweighting procedure resulted in a highly significant logistic regression coefficient to measure the influence of a positive VAA advice on the vote ($t = 10.0$, $p < 0.0001$), corresponding to a *potential* effect of VAA advice of 16%. The test shows to be robust for

differences in balanced reweighting schemes (means only, also variances, also correlations).

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